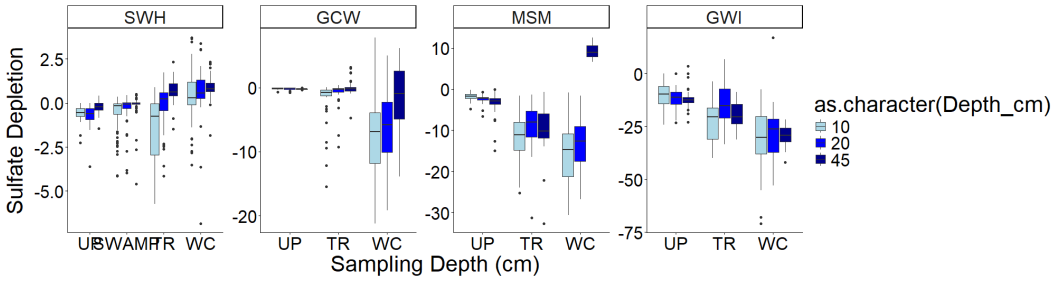
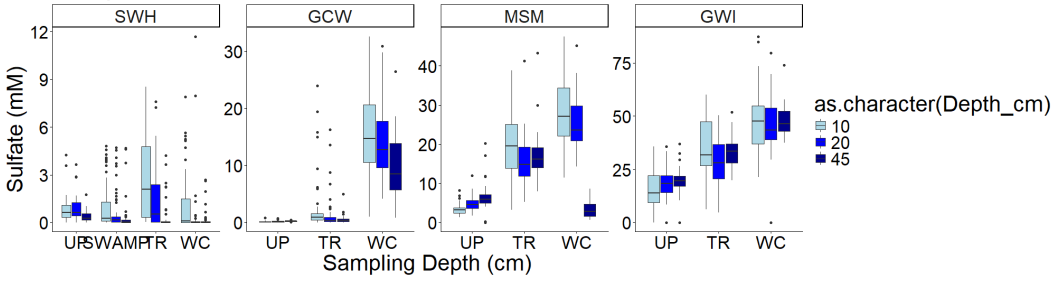
GCReW Symposium Poster Ideas

- Take a look at synoptic site porewater data and look through the years to see if we are observing increases in specific analytes through time in the transition zones compared to uplands/wetlands.
- Specifically looking at SO4, Cl, H2S, and DOC to see if we are seeing salinization and subsequent DOC release
- Analyses to look at variation through space and time. This would be comparing depths to see if there are significant differences across 10, 20, and 45cm. Also may look at interannual variation - do we see significant differences across years? what about zones?

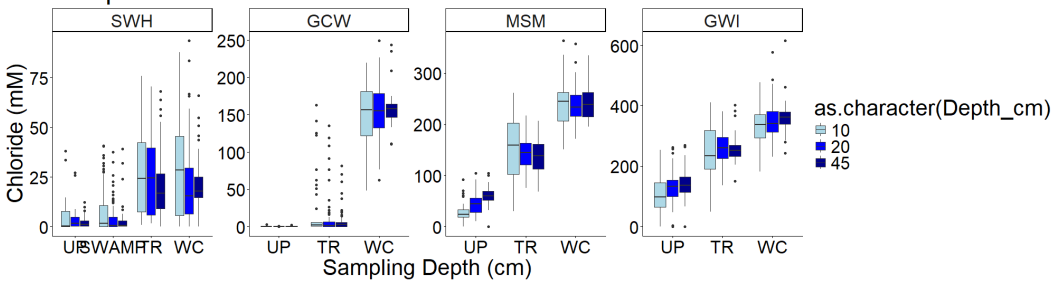
Analytes by Depth:



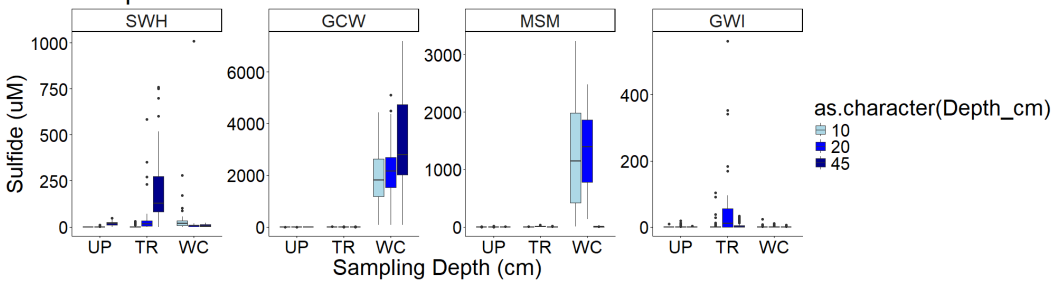
Samples: SO4



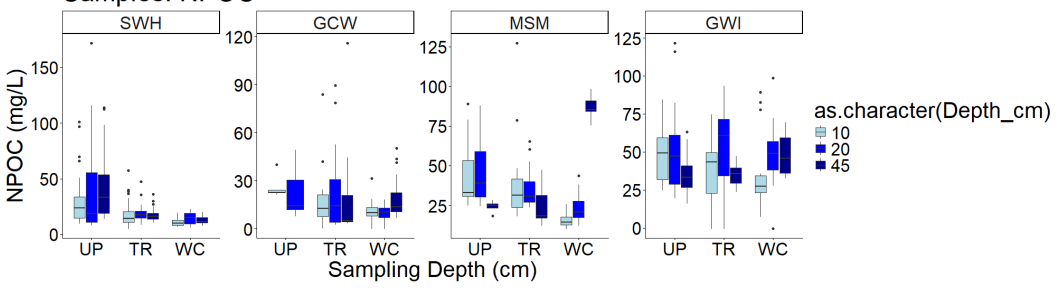
Samples: Chloride



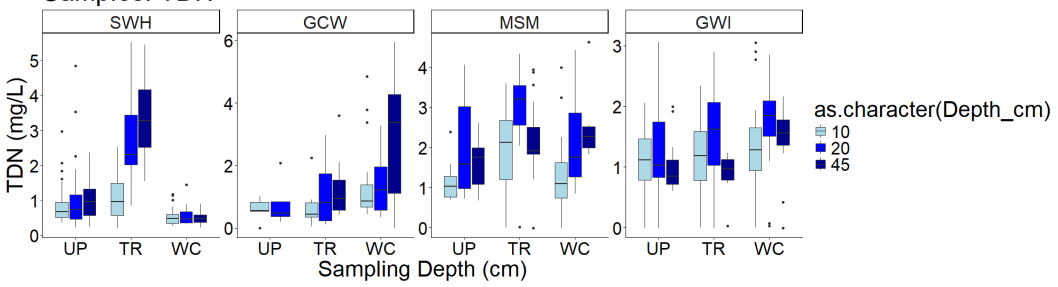
Samples: Sulfide



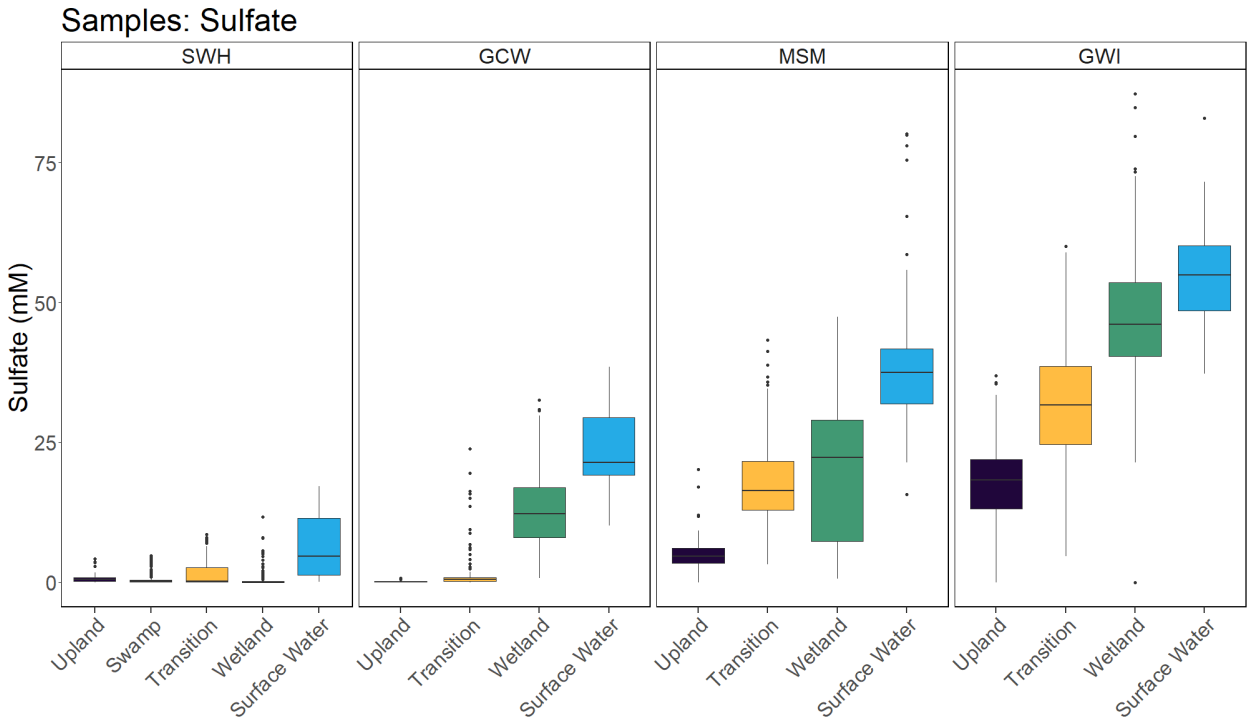
Samples: NPOC



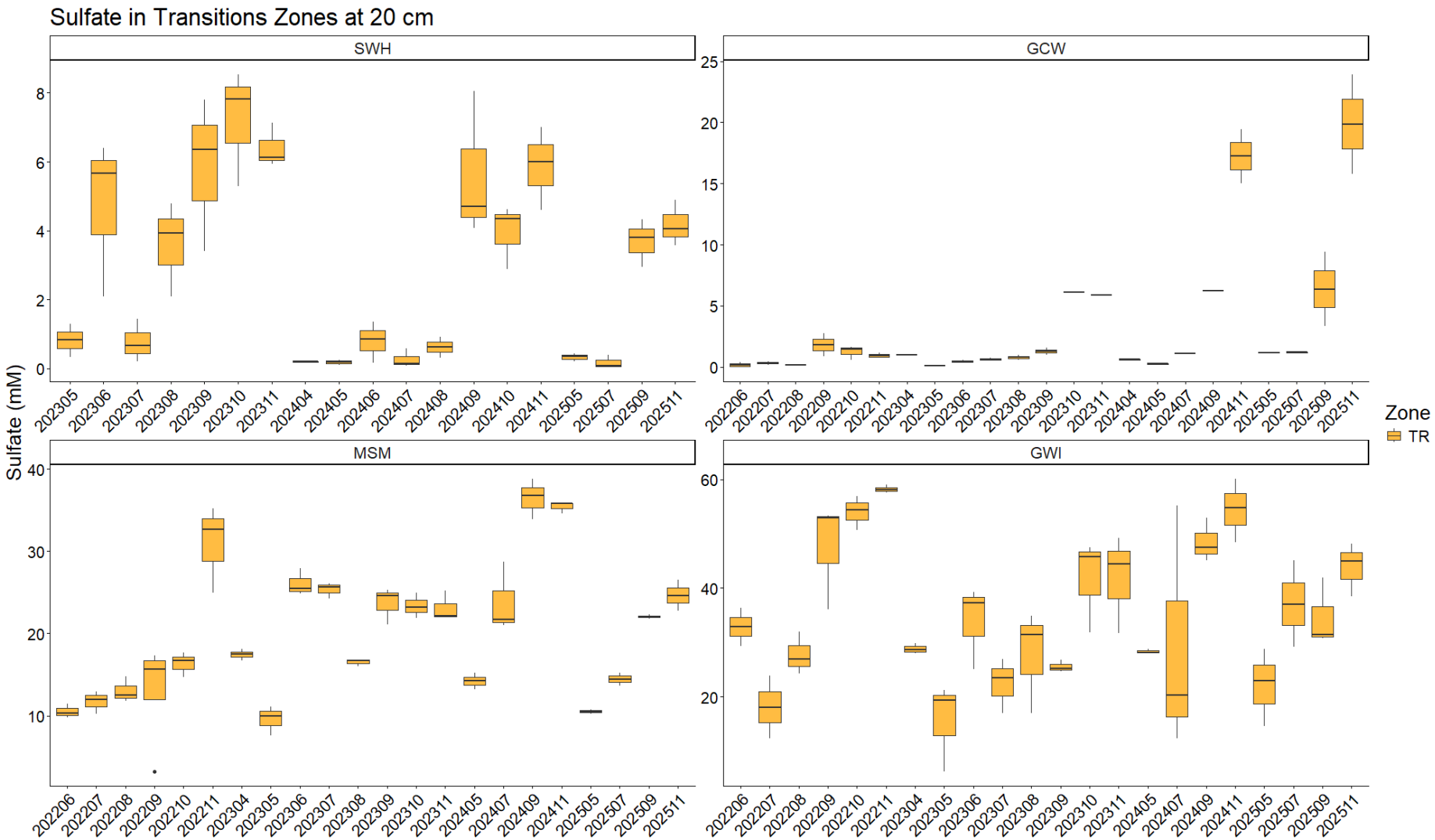
Samples: TDN



Sulfate at the various zones:

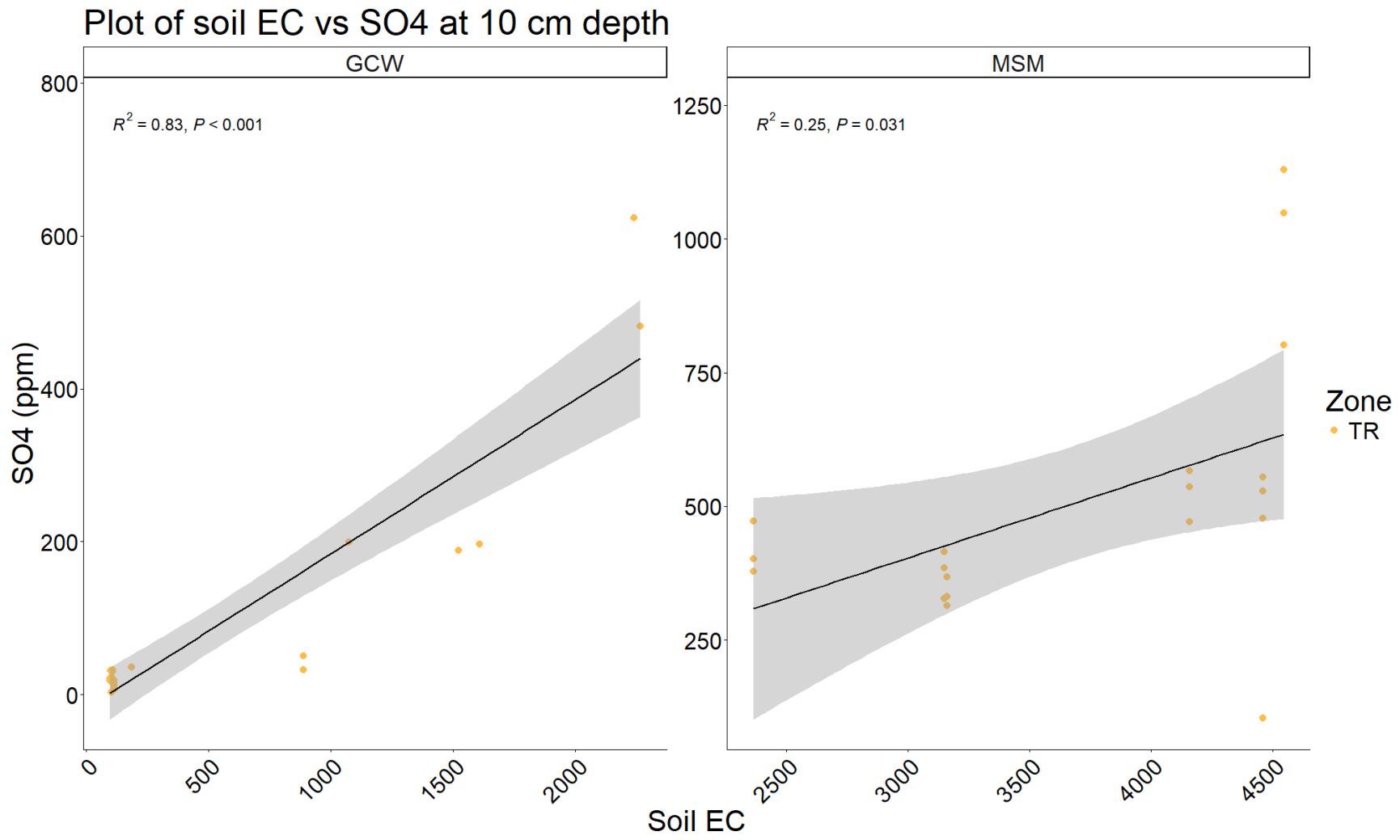


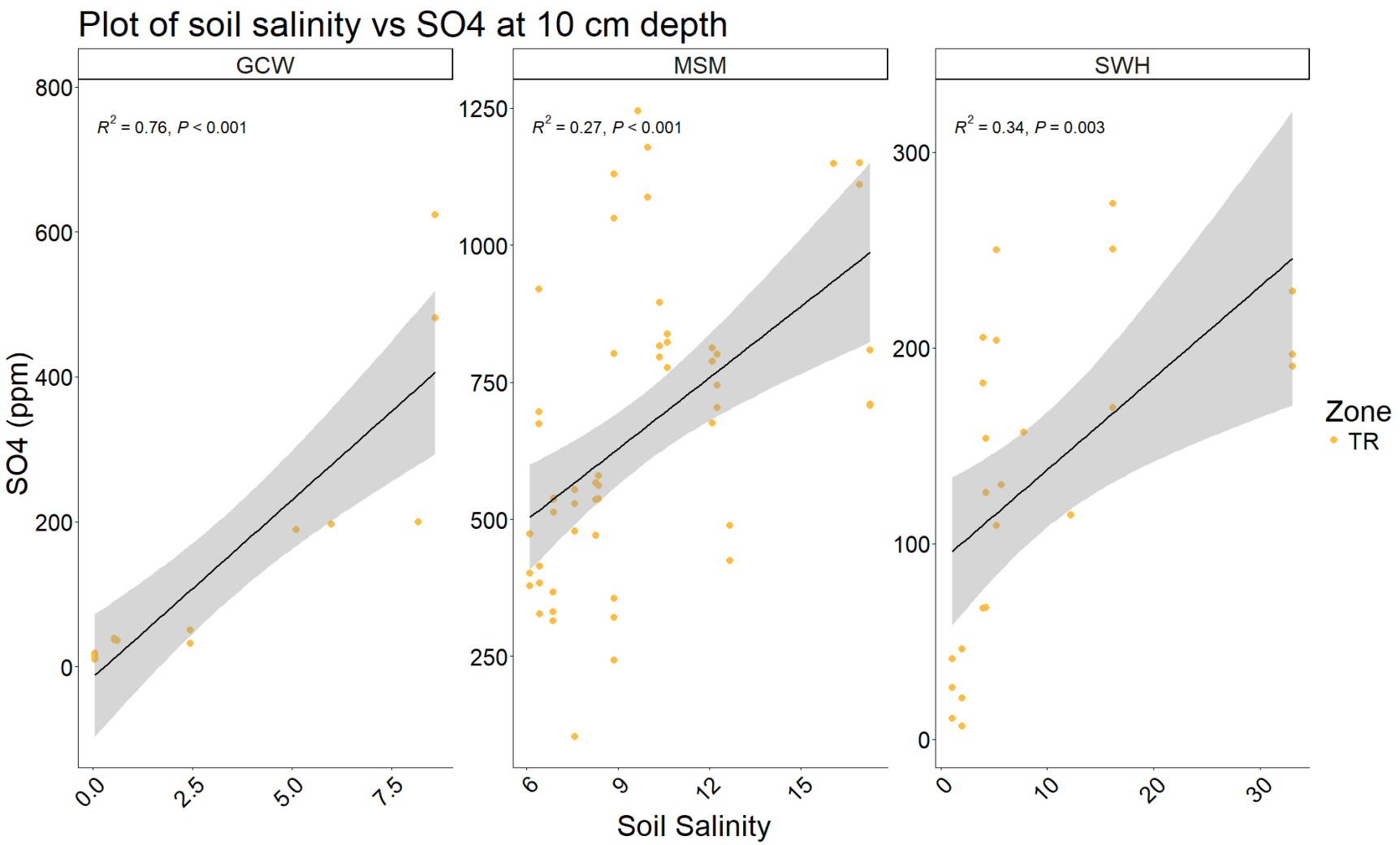
Plotted SO4/Cl through time as scatterplots/boxplots: There seems to be a seasonal trend but not so much a trend across years.

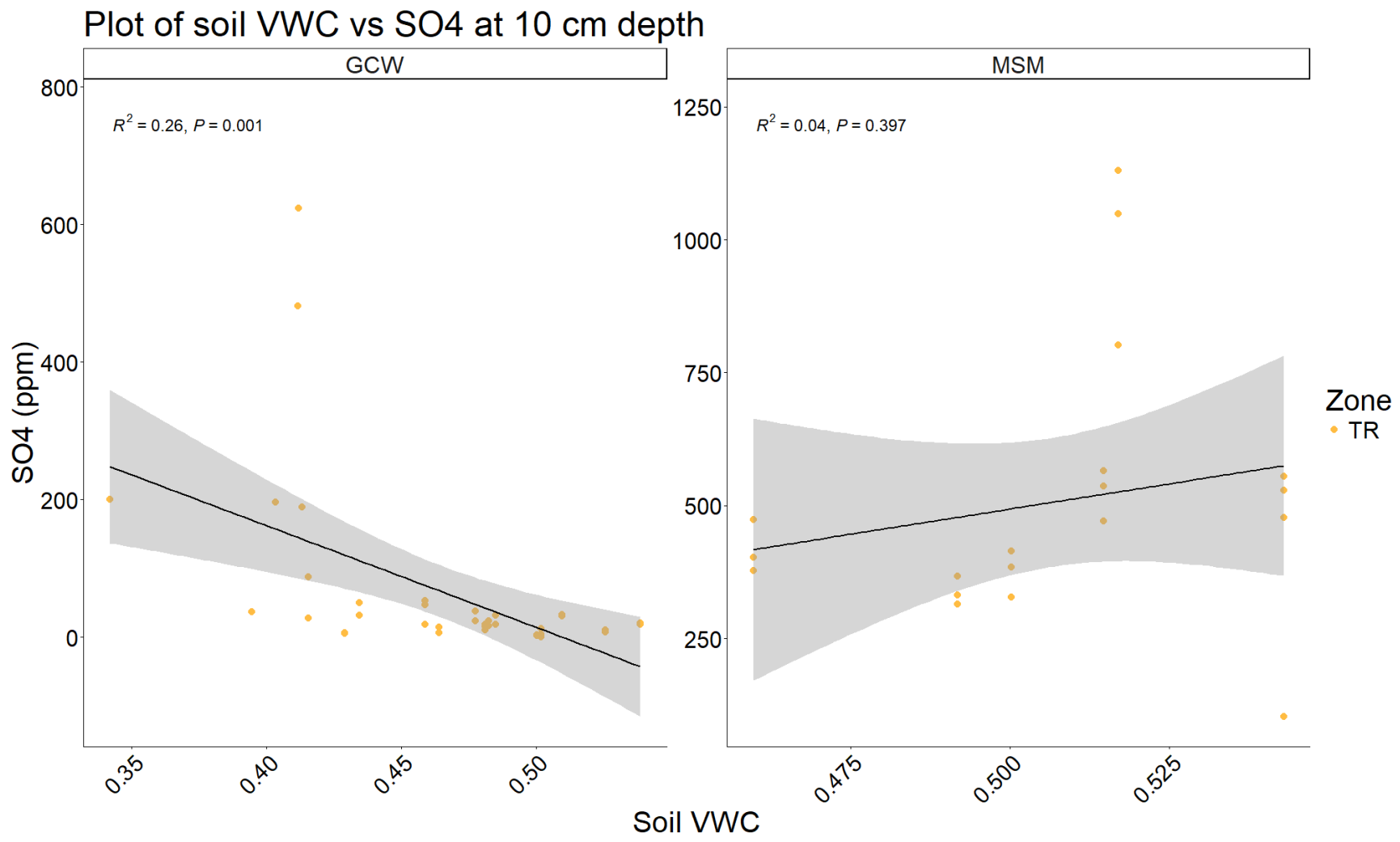


Sulfate throughout Time
SWH
GCW
MSM
GWI
Sulfate (mM)
0
2
4
6
8
0
5
10
15
20
25
10
20
30
40
20
40
60
2023-07
2024-01
2024-07
2025-01
2025-07
2023
2024
2025
2026
Zone
TR

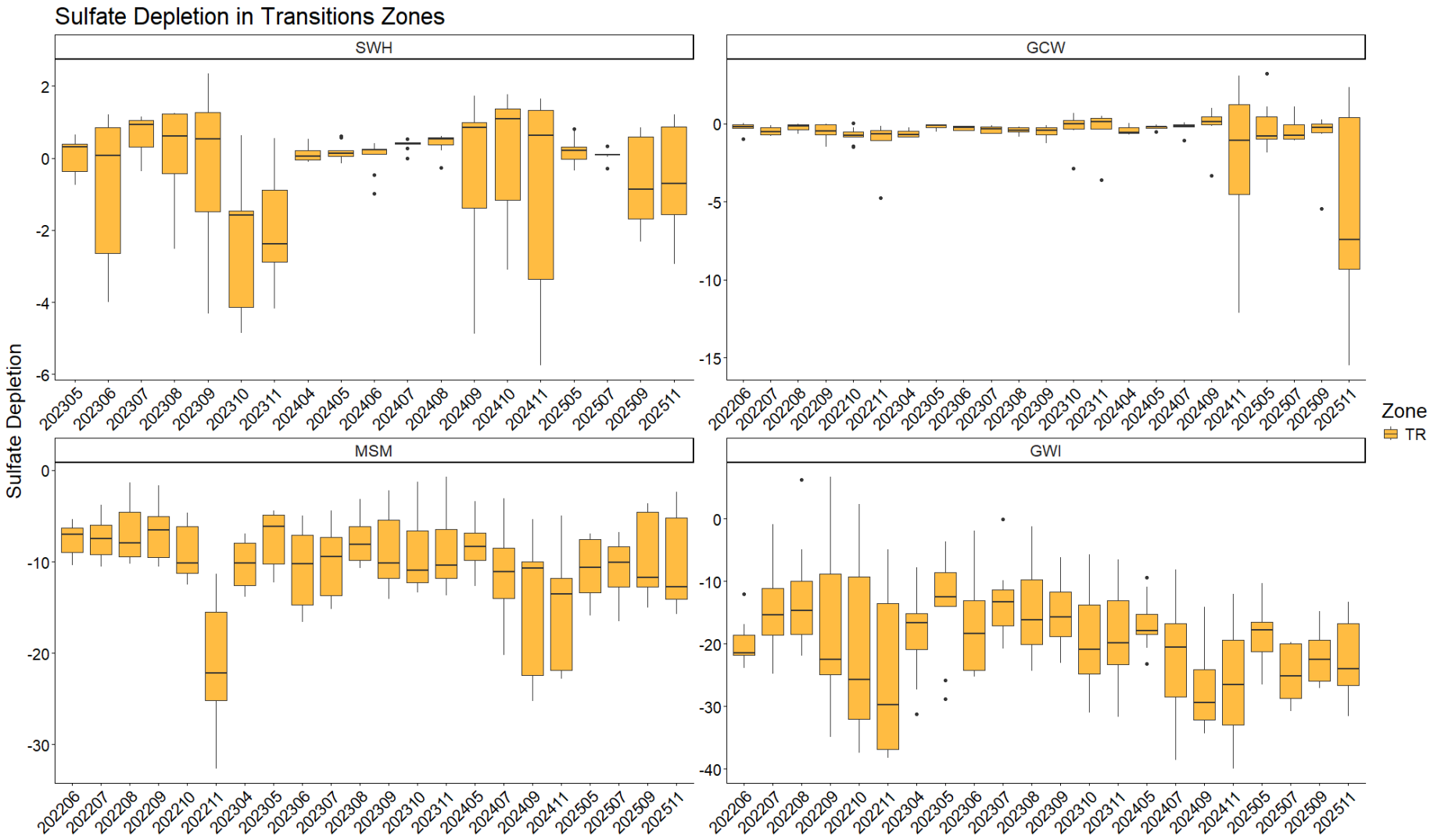
With L2 Data:

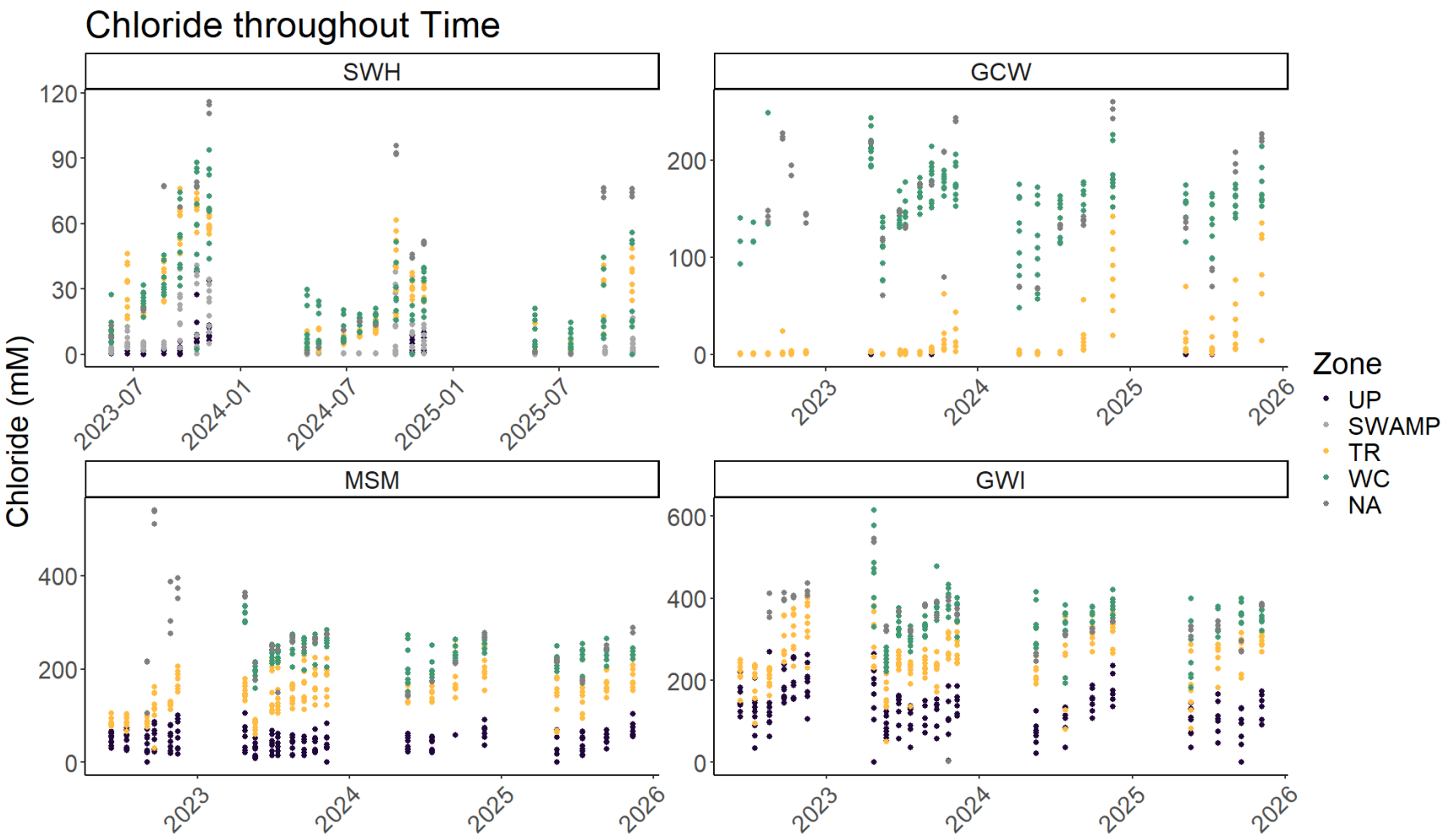

Plot of soil EC vs SO4 at 10 cm depth
GCW
MSM
R^2 = 0.83, P < 0.001
R^2 = 0.25, P = 0.031
SO4 (ppm)
Soil EC
Zone
TR
800
600
400
200
0
1250
1000
750
500
250
0
500
1000
1500
2000
2500
3000
3500
4000
4500

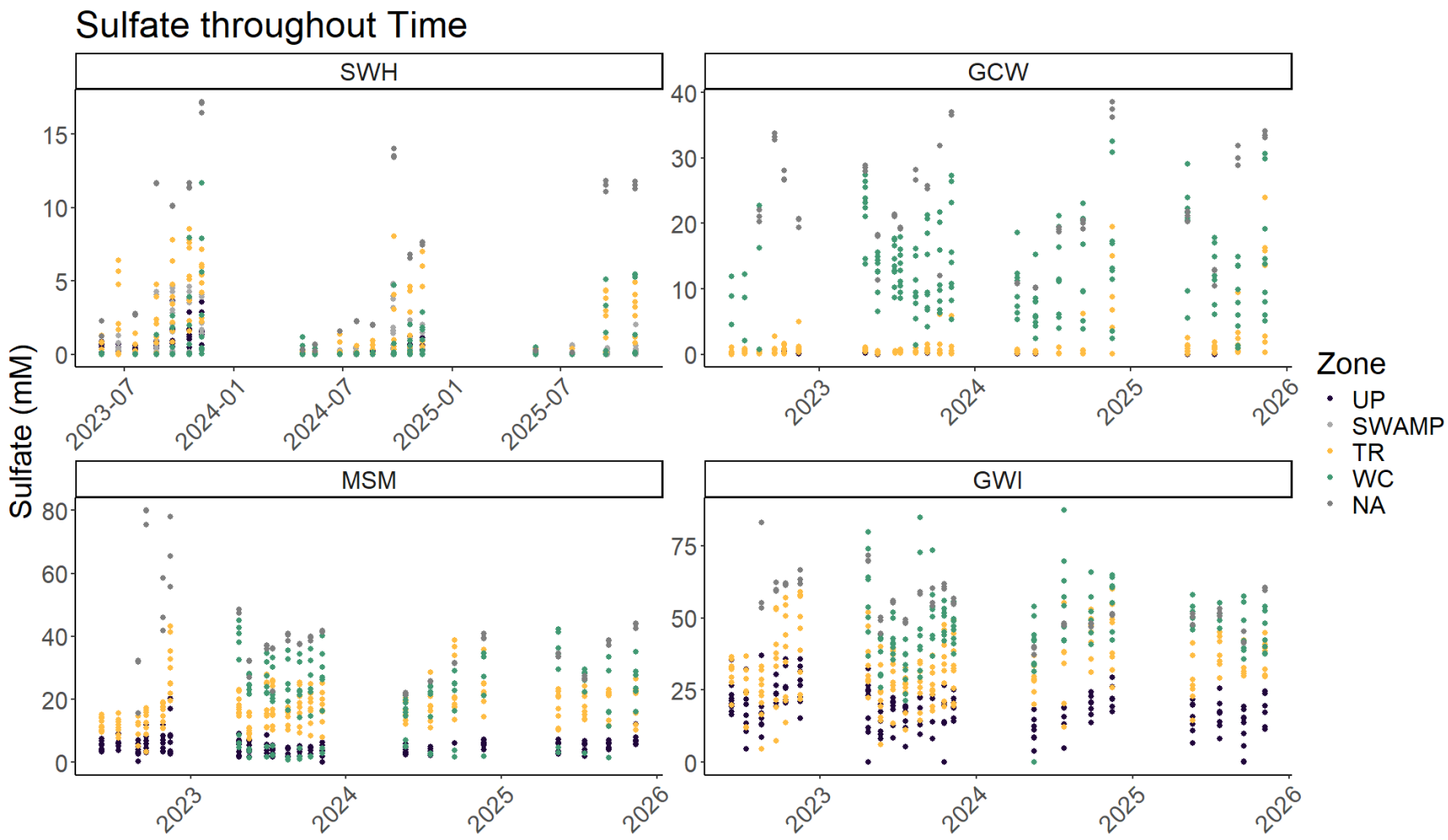

Plot of soil salinity vs SO4 at 10 cm depth
GCW
MSM
SWH
R^2 = 0.76, P < 0.001
R^2 = 0.27, P < 0.001
R^2 = 0.34, P = 0.003
SO4 (ppm)
Soil Salinity
Zone
TR



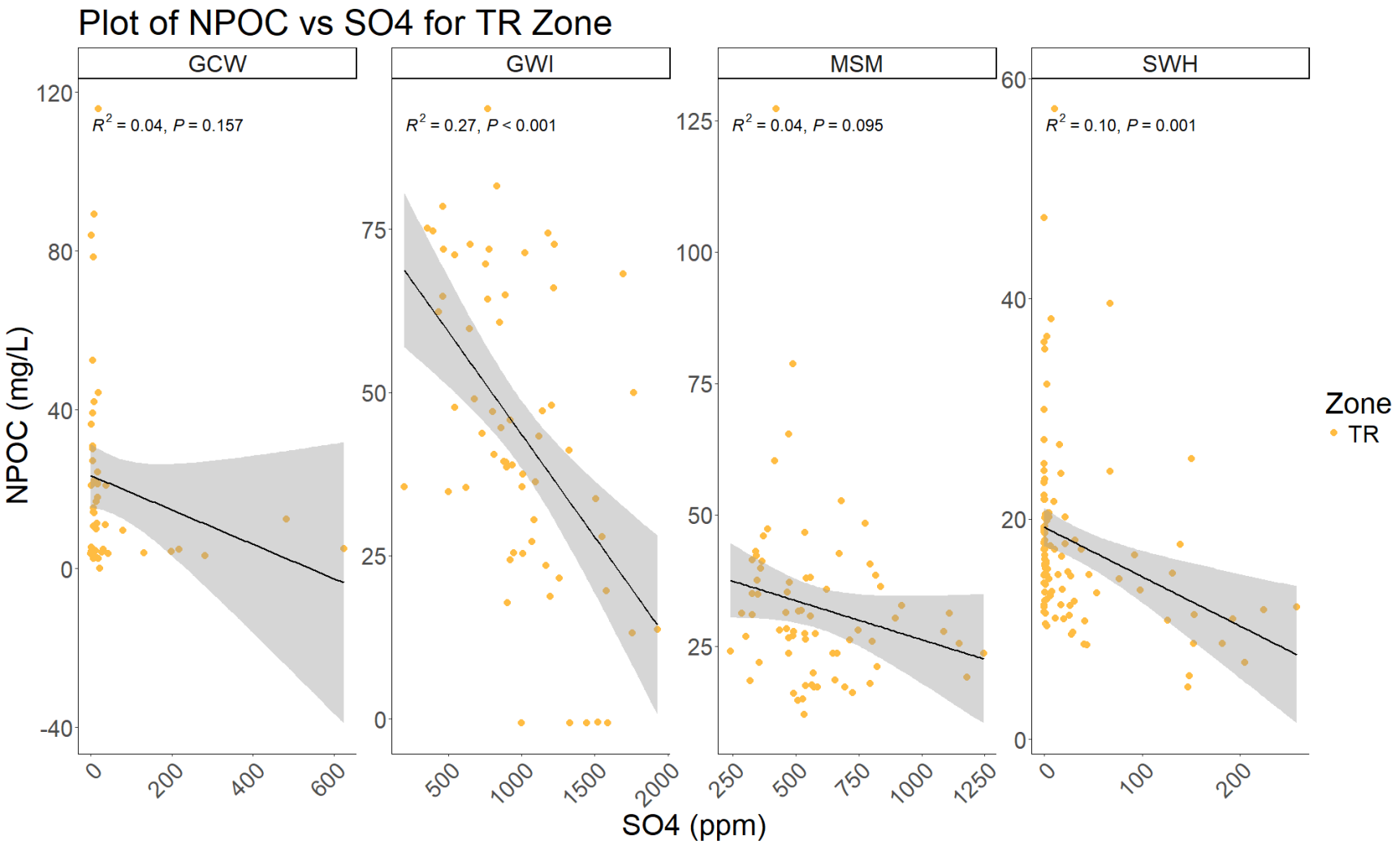
Plotted Sulfate depletion through time:

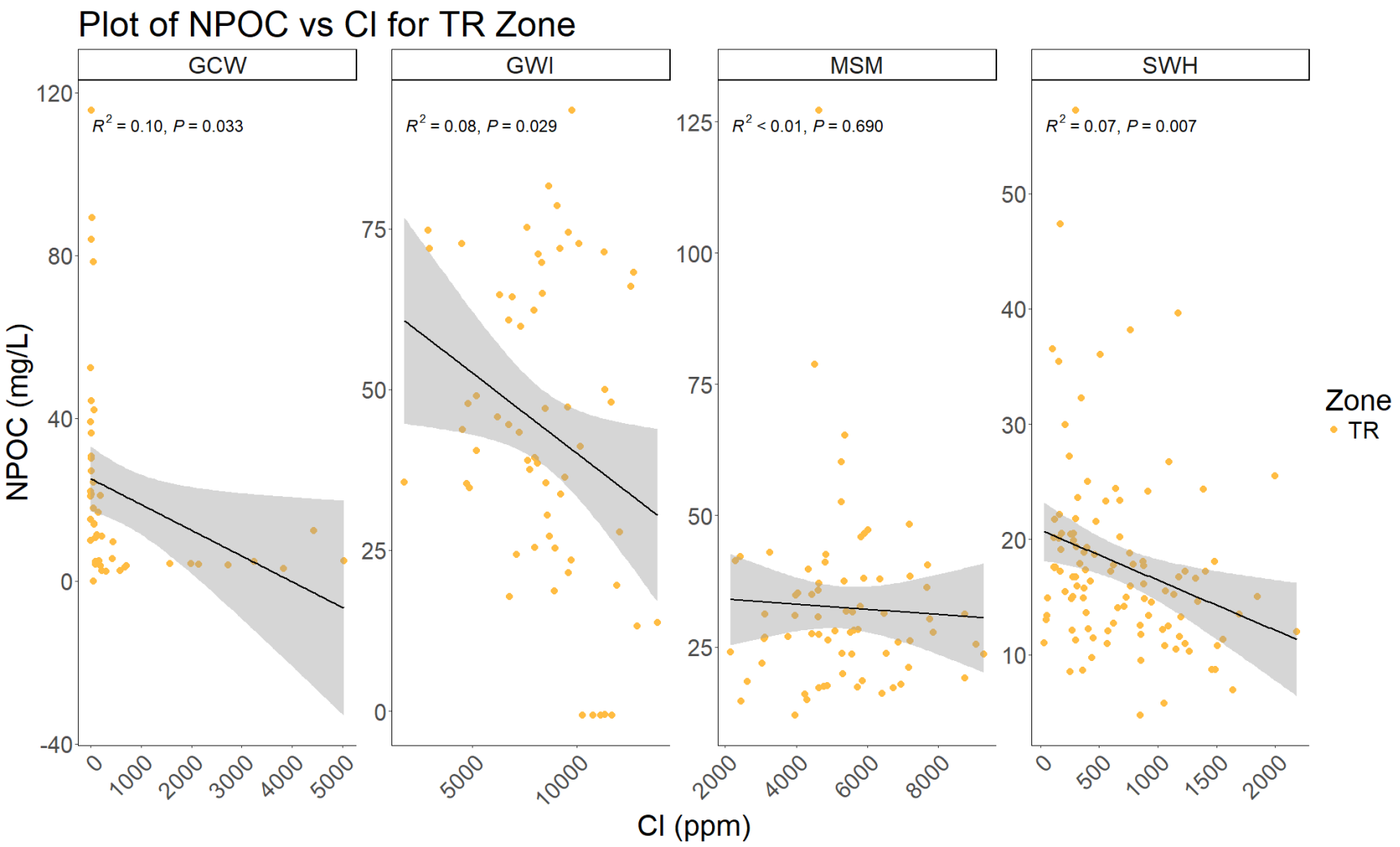

Sulfate Depletion in Transitions Zones
SWH
GCW
MSM
GWI
Sulfate Depletion
Zone
TR


Chloride throughout Time
SWH
GCW
MSM
GWI
Chloride (mM)
120
90
60
30
0
200
100
400
600
2023-07
2024-01
2024-07
2025-01
2025-07
2023
2024
2025
2026
Zone
UP
SWAMP
TR
WC
NA

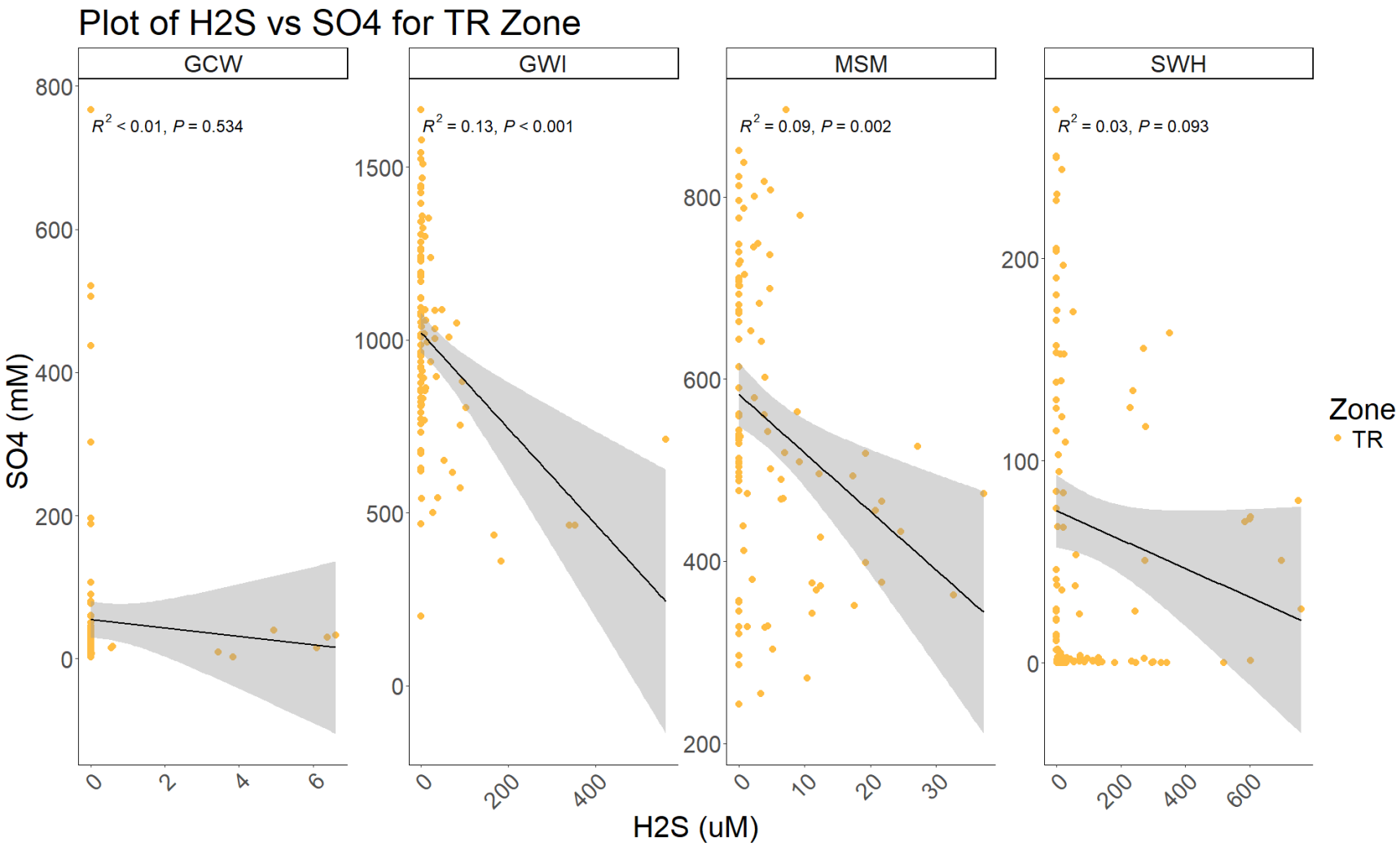

Sulfate throughout Time
SWH
GCW
MSM
GWI
Sulfate (mM)
Zone
UP
SWAMP
TR
WC
NA
2023-07
2024-01
2024-07
2025-01
2025-07
2023
2024
2025
2026

DOC vs SO4/Cl for only Transition Zones: Looks like a generally inverse relationship.

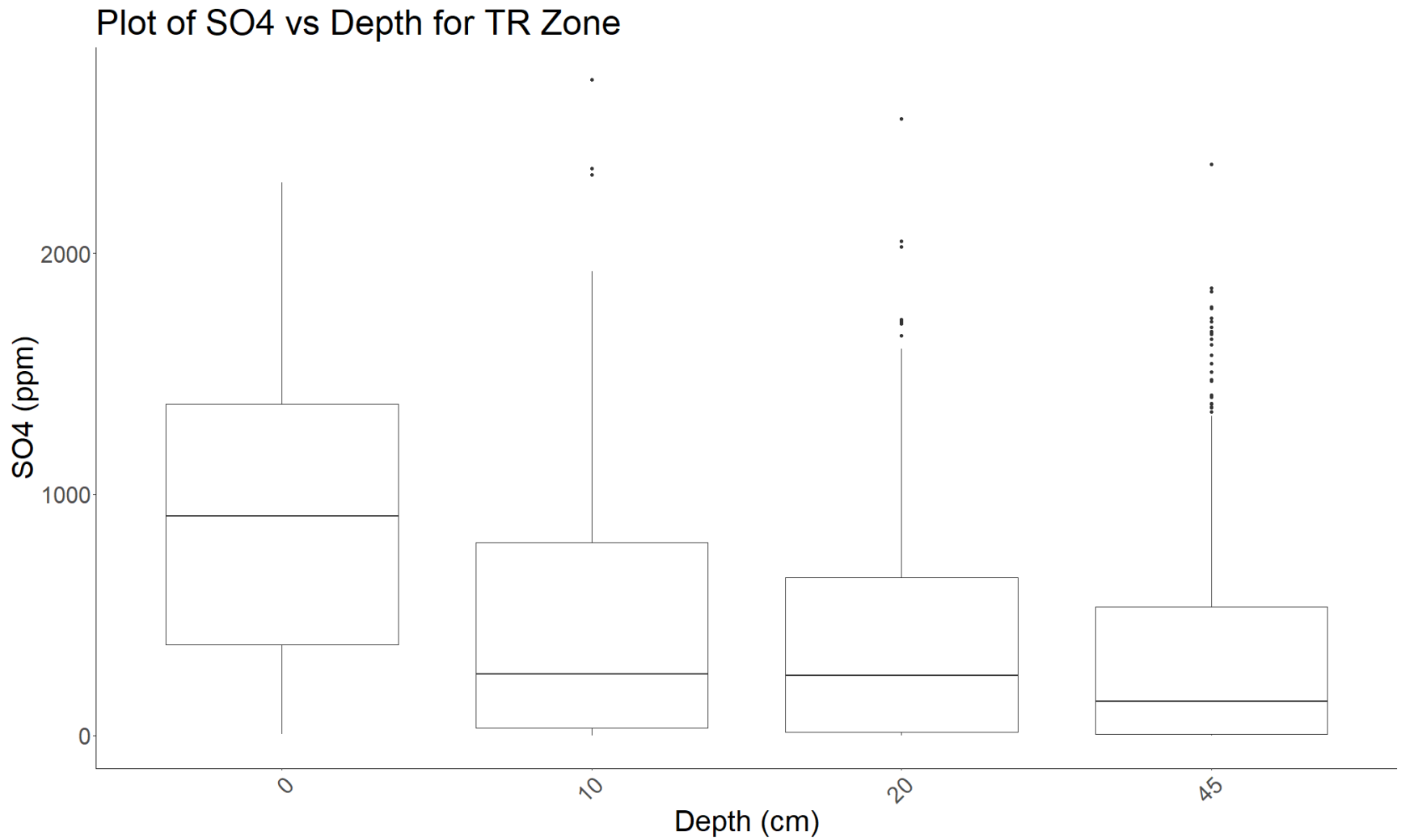


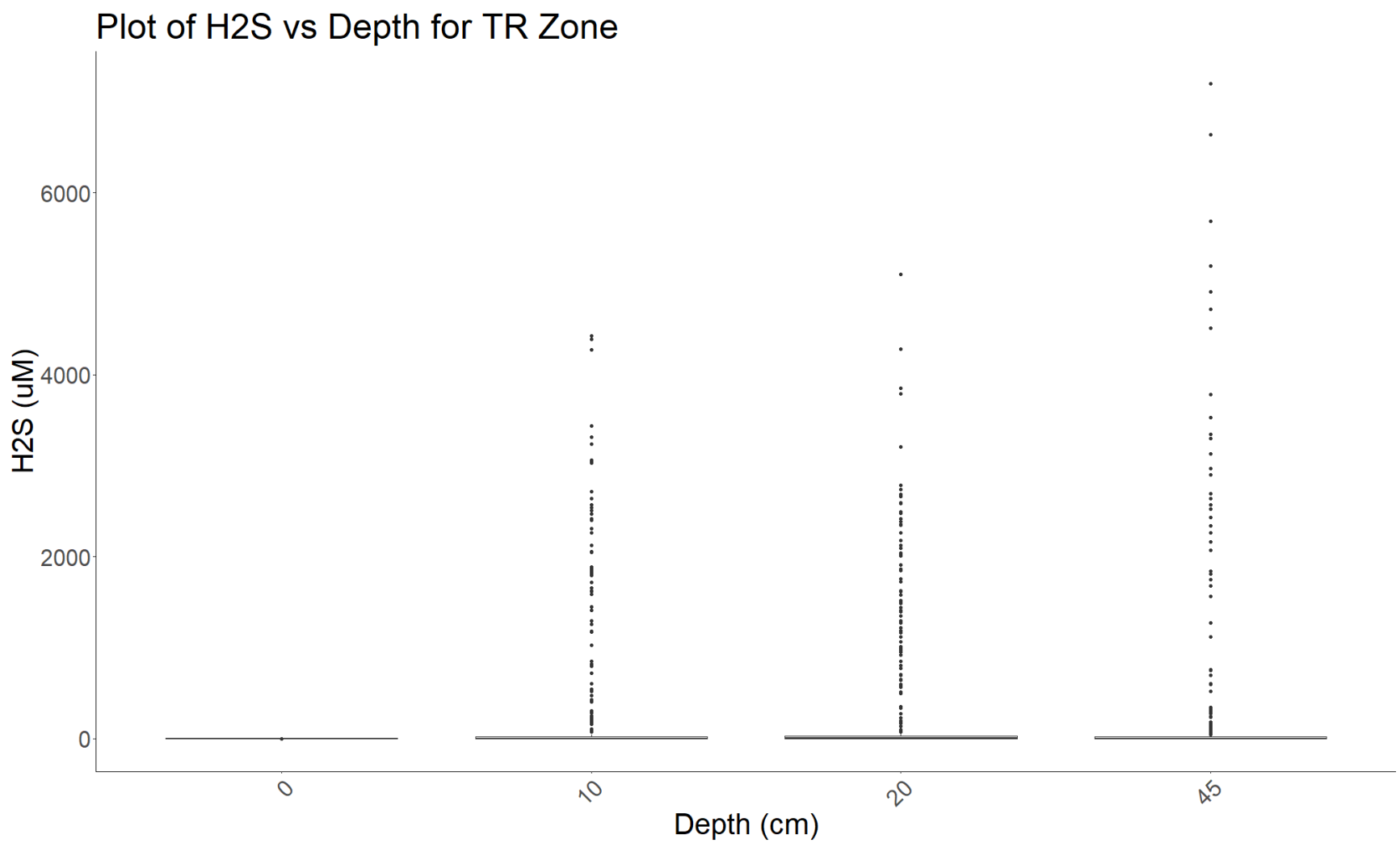

Plot of NPOC vs Cl for TR Zone
GCW
GWI
MSM
SWH
R^2 = 0.10, P = 0.033
R^2 = 0.08, P = 0.029
R^2 < 0.01, P = 0.690
R^2 = 0.07, P = 0.007
NPOC (mg/L)
Cl (ppm)
Zone
TR

Graphed SO4 vs H2S in Transition Zones: It seems to be a generally inverse relationship, as expected.

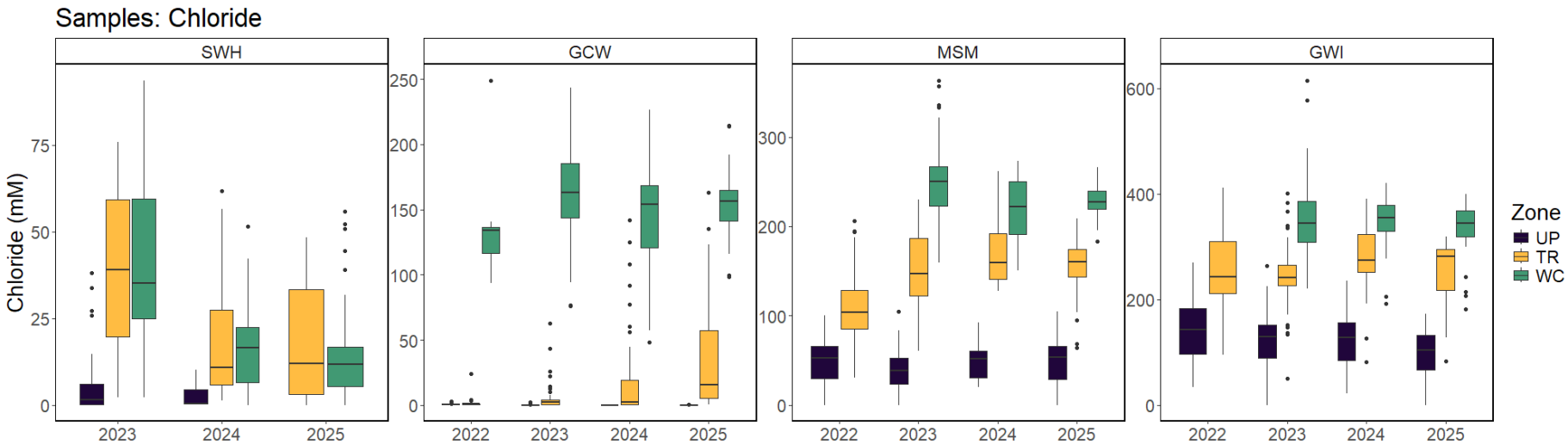


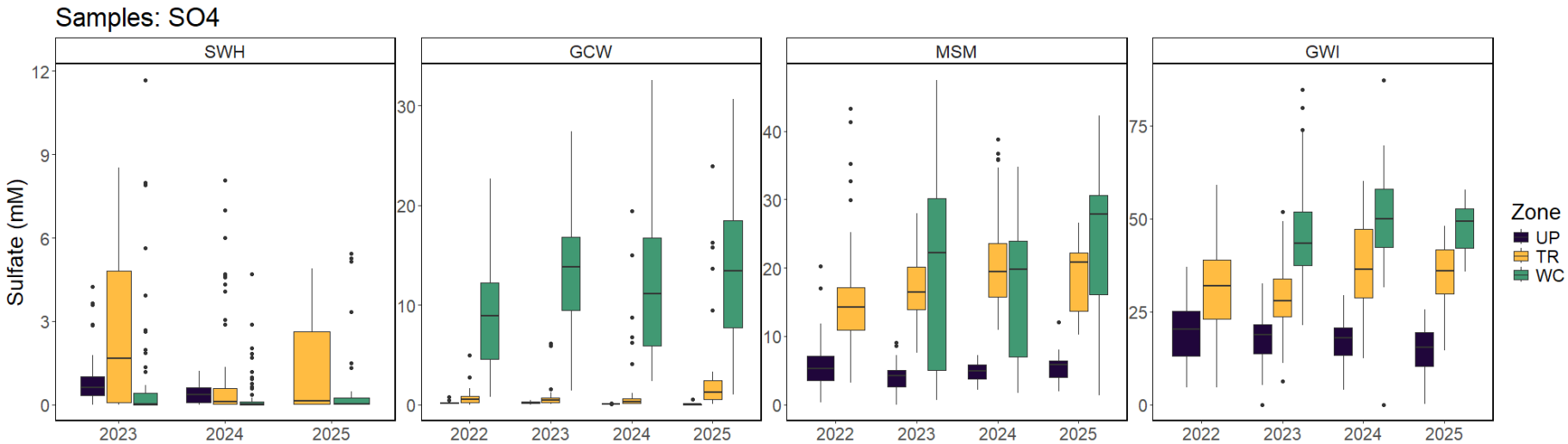
H2S and SO4 vs Depth: Looks like SO4 decreases and H2S increases with depth.



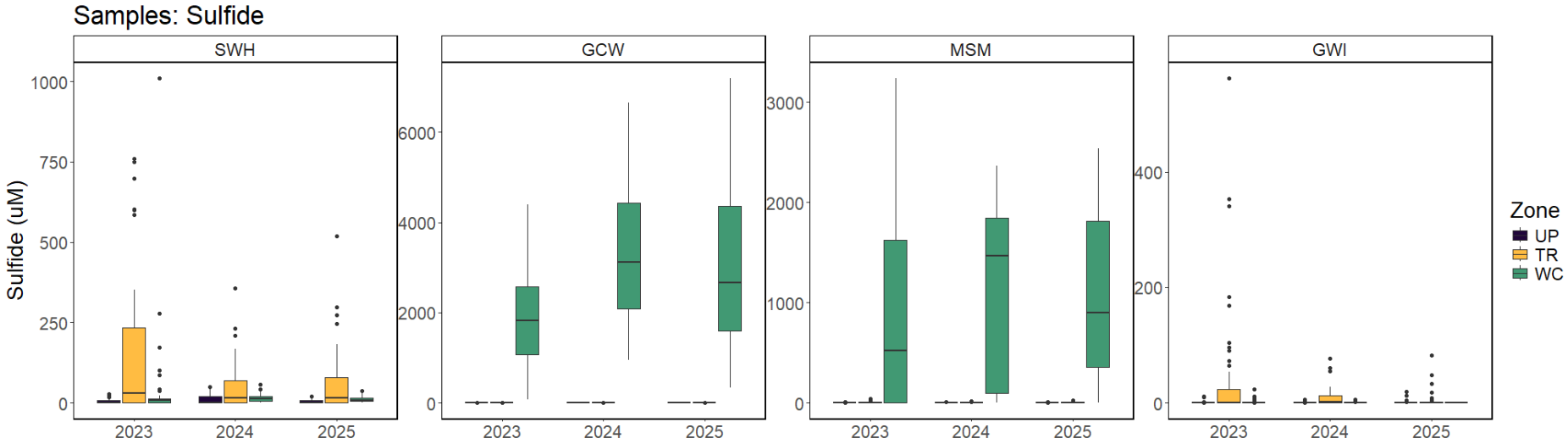

Plot of H2S vs Depth for TR Zone
H2S (uM)
6000
4000
2000
0
0
10
20
45
Depth (cm)

It looks like maybe SO4/Cl is increasing in the TR at GCW? Missing SWH UP 2025 SO4/Cl data?

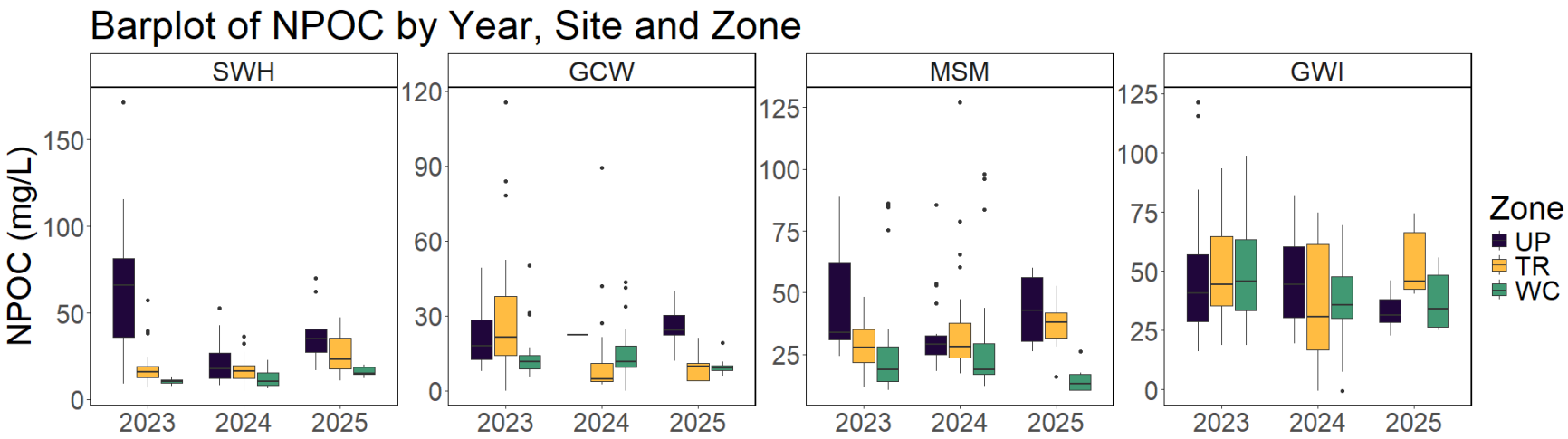


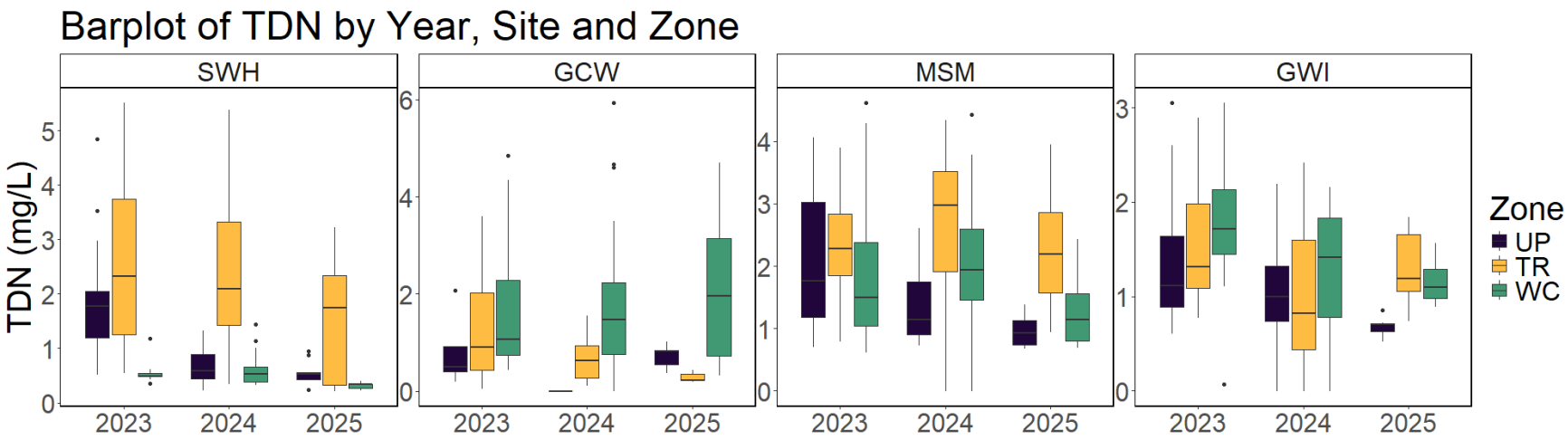


No pattern in H2S data?

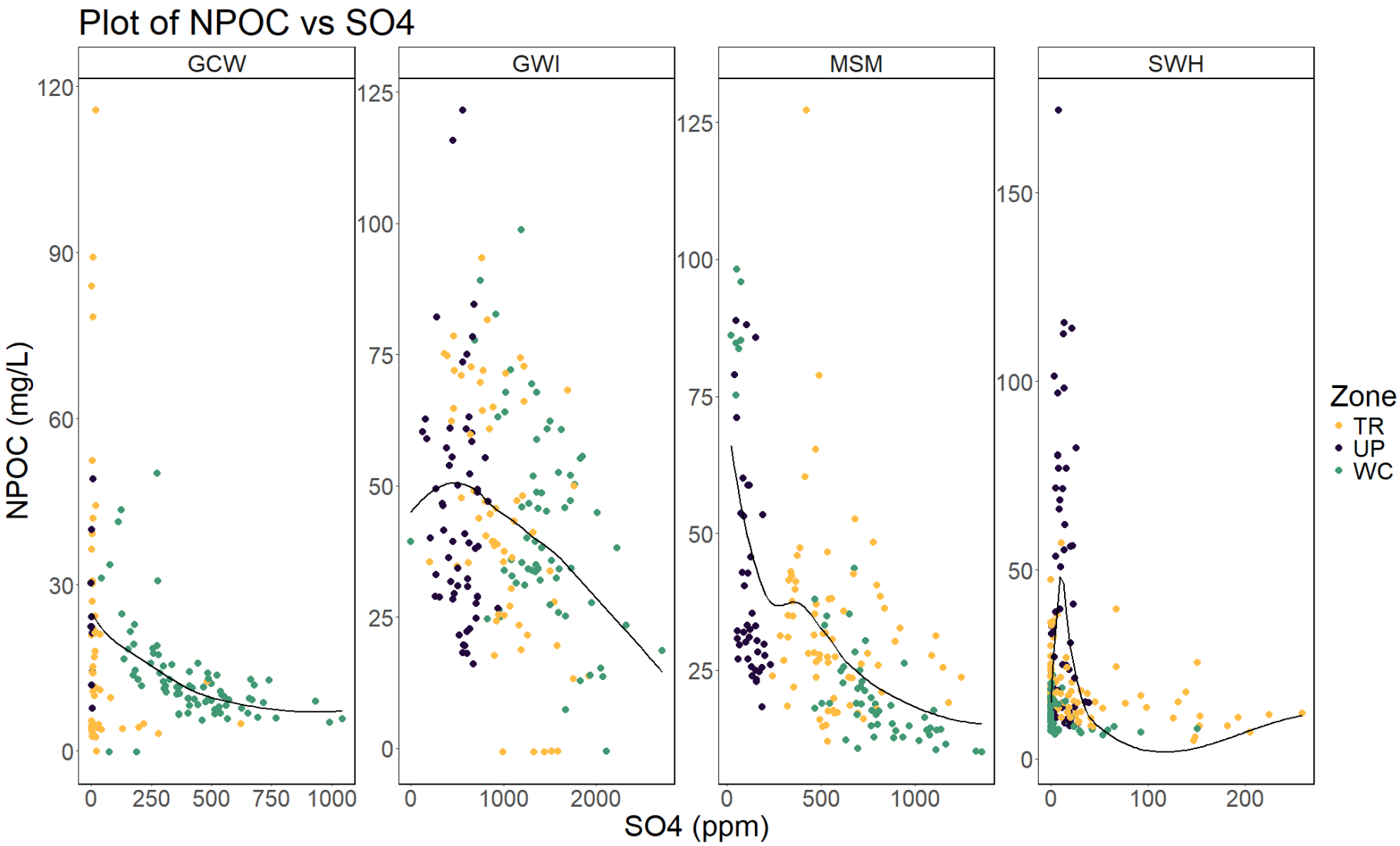


No pattern in DOC data?

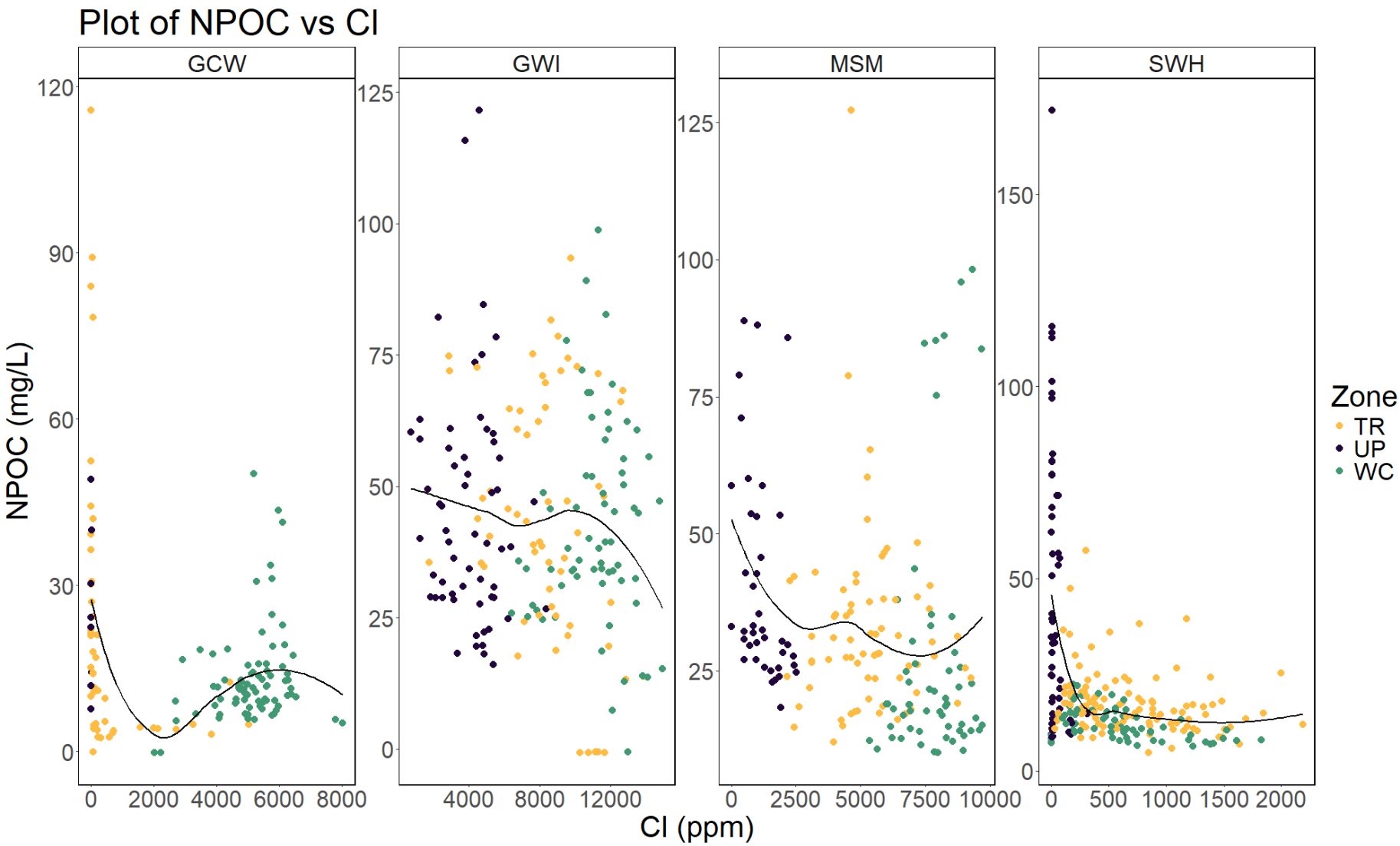
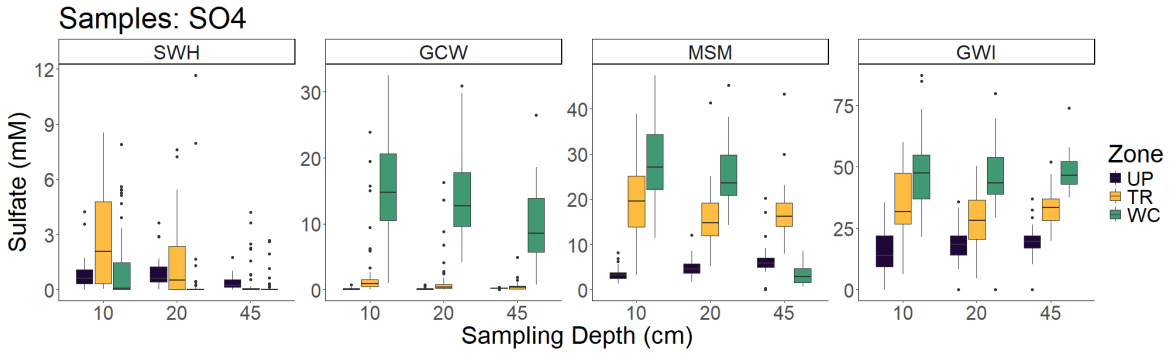
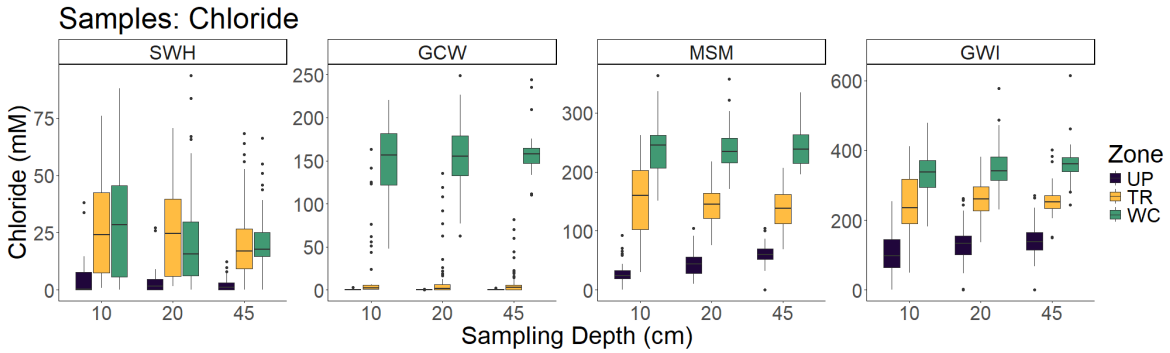
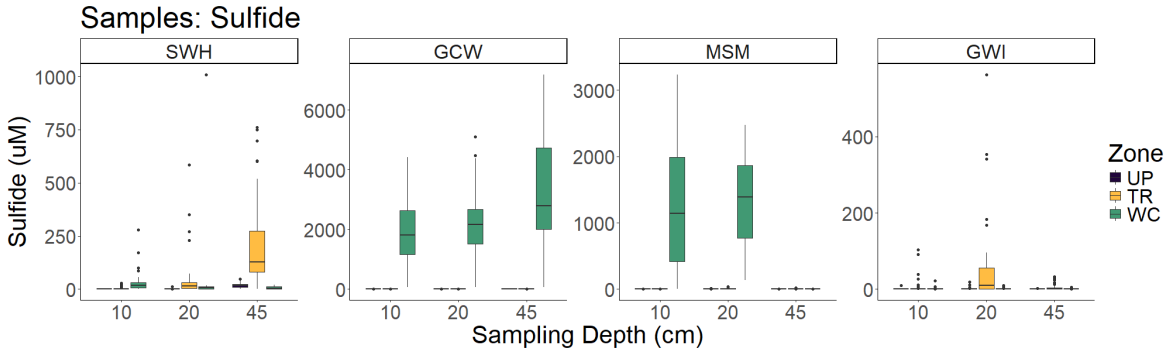
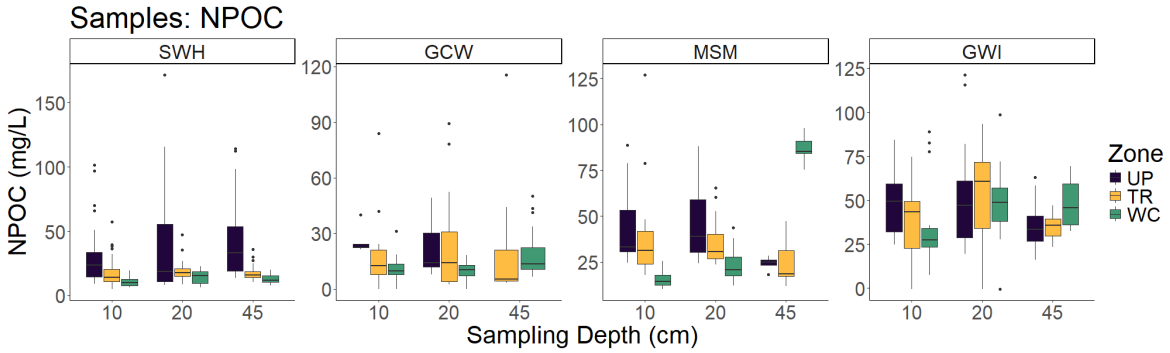
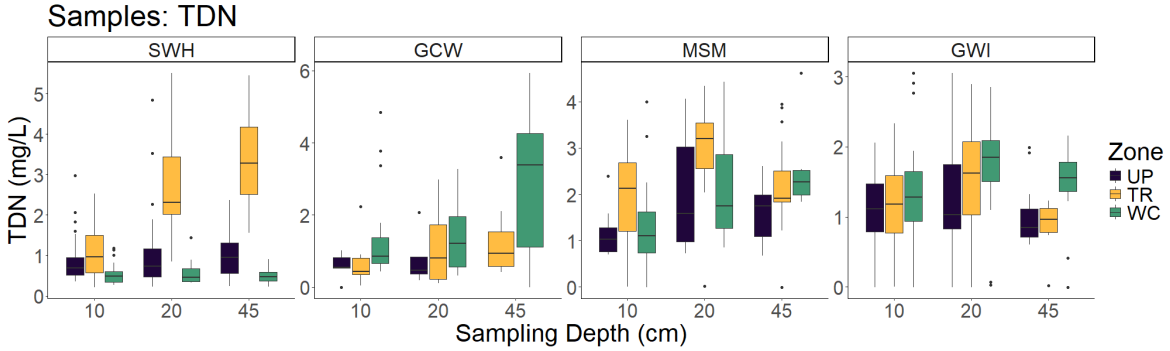




DOC decreases as SO4 increases across Zones?



No clear pattern in NPOC vs Cl?



Sulfide, Sulfate, and TDN seem to have the largest differences by depth